

An Infrastructure Oriented for Cataloging Services and Reuse of Analysis Patterns

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Abstract. Patterns have been employed as a mechanism for reuse in several phases of software development. Analysis patterns consist of artifacts for reuse during the requirements analysis and conceptual modeling. However, they are generally, documented in a textual manner which is not precise to be treated by a computer, thus limiting the dissemination and a wider reuse. Within the geo-processing area, Spatial Data Infrastructures (SDI) has been used quite effectively as an instrument for the reuse of geospatial data and services. Based on the development of SDIs, this article proposes an Analysis Patterns Reuse Infrastructure (APRI) comprising web services and a metadata representation for the specification of analysis patterns, in order to support the cataloging and reusing of analysis patterns.

Keywords: Analysis Patterns, Reuse, Dublin Core, SOA, Spatial Data Infrastructure.

1 Introduction

Spatial Data Infrastructure (SDI) is a relevant base collection of technologies, policies and institutional arrangements that facilitate the availability of and access to spatial data. The SDI provides a basis for spatial data discovery, evaluation, and application for users and providers within all levels of government, the commercial sector, the non-profit sector, academia and by citizens in general [1]. Currently, most of SDI are based on Service-Oriented Architecture (SOA), allowing to create shared environments, distributed and interoperable based on Web Services [2].

The use of SDI allow the availability of spatial databases resulting from different providers, bringing facilities to the user to acquire new datasets, without the need to build them or convert them, generating duplicity and overcharging his work [3].

Appropriate documentation of data and services is a very important point in the development of an SDI. This documentation is produced through defined metadata from a standard metadata specification, for instance, Dublin Core [4].

Fowler [5] describes analysis patterns as an idea that has proven to be useful in a given practical context and that can be useful in other contexts. This category of patterns is focused on the reuse of artifacts generated during the requirements analysis and conceptual modeling phases.

Analysis patterns were a well-researched topic in the past, resulting in publications in various domains. However, recently the number of publications in this subject decreased. The work of Blaimer, Bortfeldt and Pankratz [6] places the theme back in focus, wherein the authors present a detailed review of the literature about analysis patterns and propose new research challenges in this subject.

Documentation of analysis patterns is important to describe the context in which they can be reused and make possible the sharing of knowledge among designers [7]. Describing a pattern improperly can generate redundant patterns or patterns in an inadequate context, thus compromising their reuse by other designers [8].

In the field of Geographic Information Systems (GIS), SDI has been used very efficiently as a tool for reusing services and geospatial data. In a way analogous to how geospatial data are documented and recovered in an SDI, the documentation of analysis patterns by means of metadata and its recovery made through Web services can contribute to the dissemination and increase of reuse of these patterns.

Thus, the aim of this paper is to propose a service-oriented infrastructure to support the cataloging and reuse of analysis patterns.

The rest of the paper is organized as follows: Section 2 describes related work with metadata standards and ways of specifying analysis patterns. Section 3 describes the architecture of a service-oriented infrastructure for reuse of analysis patterns. Section 4 presents some concluding remarks.

2 Related Works

2.1 Metadata Standards

Metadata consists of data about data. Metadata standards are used to standardize the set of elements of the data description. For example, a standard for geospatial metadata may contain the title, a general description, authors, the spatial limits of the data, quality aspects, and other elements, thus facilitating the search, retrieval and reuse data appropriately [4].

The structure of a data description, given by metadata, can vary according to the domain of the described data. Considering this situation, many standards have been created in order to establish structures of metadata for specific areas and thus enable a common understanding for a community of users.

The standard CSDGM (Content Standard for Digital Geospatial Metadata) is the most used standard worldwide to describe spatial data. It was created by the FGDC (Federal Geographic Data Committee) in 1994 to support construction of U.S. SDI (NSDI - National Spatial Data Infrastructure) [9]. This standard is composed by 469 elements.

The ISO 19115 standard [10] was proposed in 2003 with the aim of becoming the main standard in the field of spatial data and services. This metadata standard is very wide and rich in detail, consisting of 509 elements.

There are also generic standards metadata that can be used more widely, for example, the Dublin Core Metadata Initiative (DCMI) [11]. The Dublin Core standard is only composed of 15 elements, and usually all of these elements are present in any broader metadata standard. No element of the Dublin Core is mandatory and all of them can be repeated many times during a data description. Its simplicity brings benefits such as lower costs of implementation and promotion of interoperability between data from different domains, however, the small number of elements that make up the standard does not provide sufficient semantic descriptions of data to describe complex domains [12].

Although the Dublin Core standard is generic and very simple, it can be used as a basis for creating profiles for specific domains from the addition of new elements, specification of rules for the occurrence of the elements and definition of types of values for each element [4].

2.2 Templates for Analysis Patterns Specification

Analysis patterns can be specified from predefined templates and outlines. The usability of an analysis pattern is reduced if the author does not use a template or uses an incomplete one to specify the pattern [6].

So far, there is no widely accepted template for specifying analysis patterns, therefore, different methodologies are used for specification. Quite often they are derived from templates geared to specify design patterns and does not fully meet the specificities of analysis patterns [6].

Considering the shortcomings in the templates oriented to specification of analysis patterns, Pantoquilha, Raminhos and Araújo propose in [13] a specific template for analysis patterns documentation. This template contains the elements of the previous approaches used to describe patterns, as well as several additional elements for a more complete description of the analysis patterns.

3 A Web Services Infrastructure for Reuse of Analysis Patterns

Based on the infrastructure for the reuse of geospatial data, this paper proposes an Analysis Patterns Reuse Infrastructure (APRI), composed of Web services, thereby featuring architecture of SOA (Service-Oriented Architecture) [14].

The scenario offered by SOA has providers, integrators and service users as actors, and users can be human or client softwares. Thus, data and services of the APRI can be accessed by human users and software clients.

The APRI (Figure.1) consists of the following components:

- *Pattern Portal*: contains a set of Web sites focused on obtaining the analysis patterns and tools and services that provide discovery, cataloging and reuse of them.

- *Metadata Repository*: are repositories that contain metadata in XML (eXtensible Markup Language) for the specification of analysis patterns and services contained in the APRI. The template used for the specification of analysis patterns is a customization of the 15 elements of the Dublin Core standard with the items of the template proposed by Pantoquillo, Raminhos and Araújo [13].
- *Analysis Pattern Repository*: are repositories that contain the analysis patterns in the XMI format (XML Metadata Interchange), allowing their use for visualization and collaboration services.
- *Portrayal Service*: are services that support the visualization of the analysis patterns of APRI.
- *Catalog Service*: these services enable the discovery and use of analysis patterns and services of APRI, based on their metadata.
- *Access Service*: these services allow to access and download the analysis patterns.
- *Collaboration Service*: these services allow the experienced designer to contribute improving the analysis patterns.

The definition of this services infrastructure for reuse of analysis patterns is based on components proposed by Béjar [15], for creating SDI.

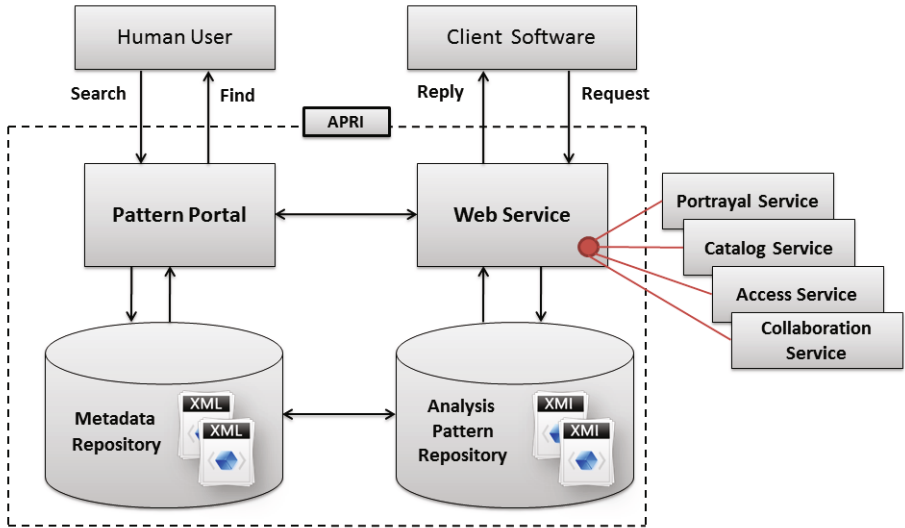


Fig. 1. Analysis Patterns Reuse Infrastructure (APRI)

4 Final Considerations

The architecture proposed in this paper allows the development of a service-oriented infrastructure for the cataloging and reusing of analysis patterns. This proposal seeks to solve a problem in the reuse of analysis patterns, related to

the manner through which patterns are documented. With a more suitable documentation, analysis patterns can be more easily discovered, studied and reused.

The Dublin Core standard has being widely used for items documentation such as work of art, museum pieces, maps, websites, etc. One of the advantages of this standard is that the metadata can be easily found by Web search engines, as they are structured by tags, making easy the semantic contextualization. Thus, the Dublin Core can also be adapted to document analysis patterns.

The use of Web services allows as much the reuse of analysis patterns by human designers as the automatic reuse done with CASE tools support. A APRI prototype is being developed by the information systems research group of *Departamento de Informática at Universidade Federal de Viçosa*.

As future work, we can mention the studies on the use of the Dublin Core as a metadata standard for documenting other computational artifacts.

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